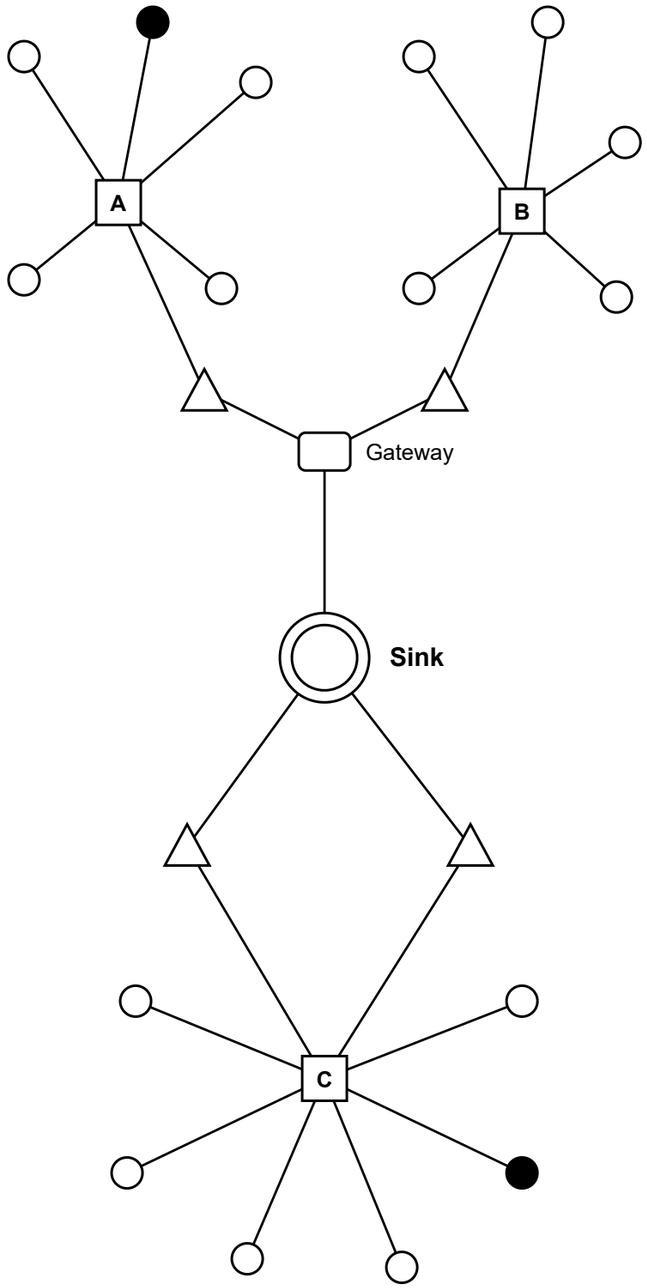
Industrial WSN Layer



Sensor Cluster Head Relay

Malicious node (Blackhole, Grayhole, Sybil, Wormhole)

Any node in radio range can forward packets

Packet statistics

Trust Management Module

Behaviour Monitoring
(packet drop, delay, identity)

Weighted Trust Evaluation (WTE)
Penalty and trust score update

Trust score
< 0.40 ?

No

Yes

Yes

Secure Routing and Recovery Module

Node Isolation (Quarantine)
Removed from routing pool

Remediation
Attack behaviour cleared after 3 rounds

Probation (Second-Chance Redemption)
Trust + 0.08 per round

Readmission
When trust score ≥ 0.65

Trust-Aware AODV (TA-AODV)
RREQ / RREP over trusted nodes in radio range

Secure routes

Status

Event-Driven Blockchain Layer

Detection Event Record

Merkle Root Generation
(SHA-256)

Proof-of-Work Mining
(Difficulty 2)

Block Sealed by Sink Node
One block per detection round

Tamper-Proof Ledger
Hash chain integrity check

Ledger data

Monitoring Layer

FastAPI Backend
Live simulation engine

WebSocket (live updates)

React Dashboard
Topology, Trust, Attacks, Ledger,
Routing and Performance